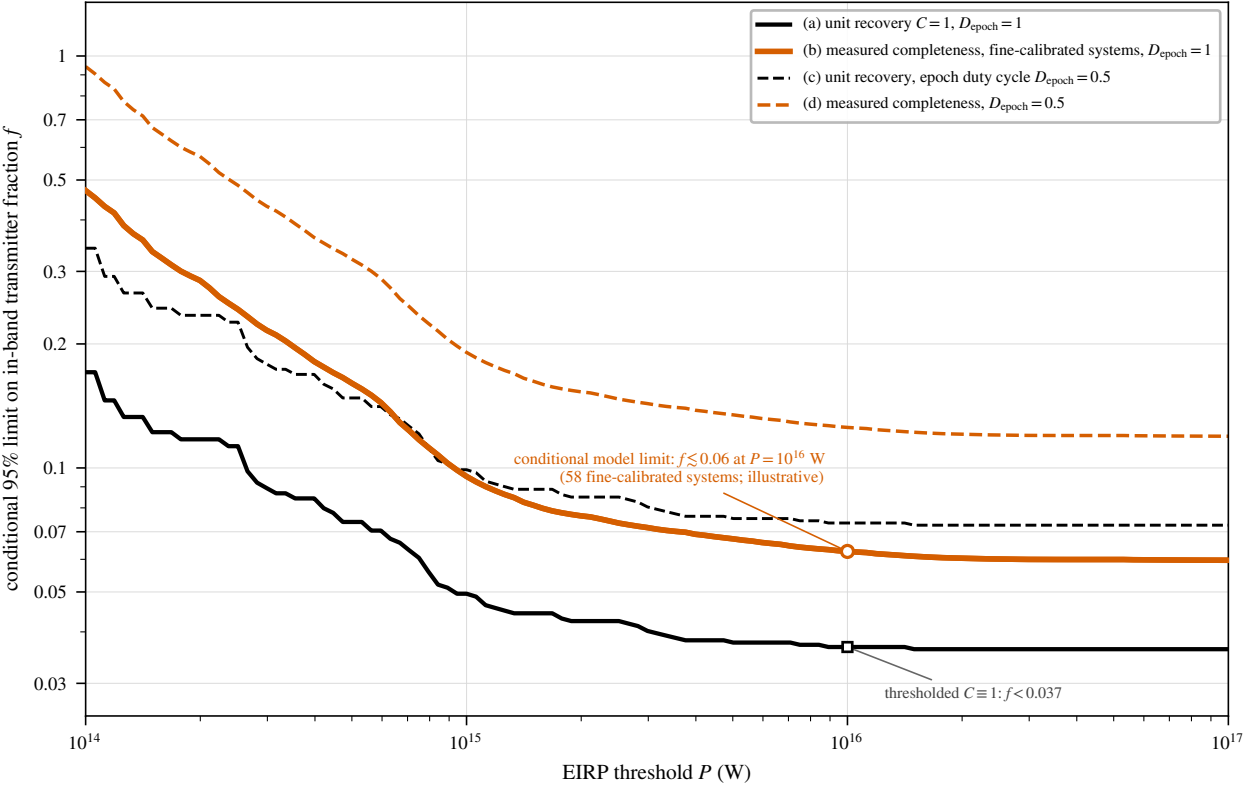




f solves $\prod_i (1 - f D_{\text{epoch}} C_i) = 0.05$, with C_i the per-system recovery at threshold P and D_{epoch} the assumed epoch duty cycle (independent Bernoulli activity across a system's execution blocks; not the in-band transmitter fraction). A low-duty-cycle population is bounded only where $D_{\text{epoch}} C_i$ stays large enough to exclude $f = 1$; as $D_{\text{epoch}} \rightarrow 0$ no bound exists at any P . Curve (b) is the conditional model limit quoted in the text, illustrating the model's behavior for fine-calibrated systems.